

MAT1001 Midterm Examination

Saturday, October 28, 2023

Time: 9:30 - 11:30 AM

Notes and Instructions

1. *No books, no notes, no dictionaries, and no calculators.*
2. *The maximum possible score of this examination is **110**.*
3. *There are **13** questions (with parts), which are worth 120 points in total. **This means that you do not have to answer all the questions in order to get the full score.***
4. *The symbol $[N]$ at the beginning of a question indicates that the question is worth N points.*
5. *Write down your solutions on the **answer book**.*
6. *Show your intermediate steps **except Questions 1, 2, and 3** — answers without intermediate steps will receive minimal (or even no) marks.*

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MAT1001 Midterm Questions

1. [10] True or False? No explanation is required.

(i) If $\lim_{x \rightarrow 0} |f(x)| = 0$, then $\lim_{x \rightarrow 0} f(x) = 0$.

(ii) If $y = (f(x))^2$ is continuous on the real line, then $y = f(x)$ is also continuous on the real line.

(iii) The graph of

$$U(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases}$$

has a vertical tangent at the point $(0, 1)$.

(iv) Suppose that $y = f(x)$ is decreasing and concave up on the real line. Then for any x , if $\Delta x = dx > 0$, then $|\Delta y| < |dy|$.

(v) If f is continuous on (a, b) then f attains an absolute maximum value $f(c)$ and an absolute minimum value $f(d)$ at some numbers c and d in (a, b) .

2. [9] For each part of this question, there is only one correct answer. Choose the correct answer. No explanation is required.

(i) Consider the function $y = f(x)$ defined over the interval $[0, 1]$ as follows:

$$f(x) = \begin{cases} x - x^2, & 0 < x < 1 \\ 1, & x = 0 \text{ and } x = 1 \end{cases}$$

Which of the following is correct?

- A) The function has three *absolute* maxima.
- B) The function has no local minimum and three local maxima.
- C) The function has one local *minimum* and two absolute maxima.
- D) The function has two local *minima* and one local maximum.

- (ii) Let f be twice-differentiable on $I = (a, b)$ and continuous on $[a, b]$, and let $f'' > 0$ on I . Which of the following must be correct?

- A) For any $c \in I$, the tangent line to $y = f(x)$ at c lies below the graph of $y = f(x)$.
- B) For any $c \in I$, the tangent line to $y = f(x)$ at c lies above the graph of $y = f(x)$.
- C) The graph of f lies above the secant line joining $(a, f(a))$ and $(b, f(b))$.
- D) None of the above (A), (B), (C) is true.

- (iii) Consider using Newton's method to solve the equation $f(x) = 0$ where

$$f(x) = a - (x - b)^2.$$

Here, $a > 0$ and b is arbitrary. What initial guess x_0 below will always approximate the largest root?

- A) Choose $x_0 < a$.
- B) Choose $x_0 \leq b$.
- C) Choose $x_0 > b$.
- D) Any initial guess x_0 can guarantee the convergence.

3. [15] Short questions: no explanation is required.

- (i) Calculate the derivative of $y = \sin\left(\cos\left(2t - \frac{\pi}{6}\right)\right)$ at $t = \frac{\pi}{3}$.
- (ii) Given that $y = x^2 + 7x - 5$ and $\frac{dx}{dt} = \frac{1}{3}$ when $x = 1$, find $\frac{dy}{dt}$ at $x = 1$.
- (iii) Find the linearization of the function $f(x) = \frac{1}{1-x}$ centered at $x = 0$.
- (iv) We wish to estimate the solution to the equation $x^3 - x - 5 = 0$ using Newton's method. Supposing we take the initial estimate $x_0 = 0$, find x_1 .

(v) Which of the following statements are always true? (There could be one or more answers.)

- A) If f is both left-continuous and right-continuous at $x = c$, then f is continuous at $x = c$.
- B) If f is both left-differentiable and right-differentiable at $x = c$, then f is differentiable at $x = c$.
- C) For any real numbers x and y , we have $|\cos(x) - \cos(y)| \leq |x - y|$.
- D) The function

$$f(x) = \begin{cases} (x-1) \cos\left(\frac{1}{x-1}\right), & \text{if } x \neq 1; \\ 1, & \text{if } x = 1 \end{cases}$$

has a jump discontinuity at $x = 1$.

4 [15] Evaluate the following limits. Use only methods and theories from Chapters 2, 3, or 4 in the textbook.

(i) $\lim_{x \rightarrow 0} \frac{x \cot(5x)}{\sin^2(x) \cot^2(3x)}$

(ii) $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1+x^2}}{\sqrt{1+x} - 1}$

(iii) $\lim_{x \rightarrow +\infty} (\sin \sqrt{x+1} - \sin \sqrt{x-1})$

5. [7] Find all vertical and oblique asymptotes for the function

$$f(x) = \frac{x^3 + 5x^2 - 7}{x^2 - 1}.$$

6. [7] Determine the first and second derivative functions of

$$f(t) = \begin{cases} \frac{1}{2}(t-2)^2 + 4, & \text{if } 0 \leq t < 2; \\ -\frac{1}{2}(t-2)^2 + 4, & \text{if } t \geq 2. \end{cases}$$

If you think the derivative functions are not defined at some points, explain and specify these points.

7. [8] Is the derivative of

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

continuous at $x = 0$? Is the derivative of $g(x) = xf(x)$ continuous at $x = 0$? Give reasons for your answers.

8. [5+4] Given the curve defined by the equation $x^2(2 - y) = y^3$:

(i) Find the equation of the tangent line to the curve at $(1, 1)$.

(ii) Find $\frac{d^2y}{dx^2}$ at $(1, 1)$.

9. [6] A car braked with a constant deceleration of 16 ft/s^2 (feet per second squared), producing skid marks (刹车痕) measuring 200 ft before coming to a stop. How fast was the car traveling when the brakes were first applied? Answer this question using the theory of antiderivatives.

10. [4+3+4] Consider the function $f(x) = x^4 - 4x^3$ defined on the real line.

(i) Determine the intervals where $f(x)$ is concave up and where $f(x)$ is concave down.

(ii) Determine the points of inflection of this function.

(iii) Determine the locations of all local maxima and minima.

11. [9] Consider the equation

$$x^3 - 2x + c = 0,$$

where c is a constant. Without solving the equation, determine the range of values of c for which:

(i) the equation has only one solution,

(ii) the equation has exactly two solutions, and

(iii) the equation has three solutions.

12. [7] A string of length L cm is used to form a triangle $\triangle ABC$ whose sides AB and AC are of the same length L_1 cm, where $2L_1 < L$. Find L_1 in terms of L so that the area of the triangle is maximized.

13. [2+5] Suppose the function $f(x)$ is continuous on $[0, 1]$ and twice differentiable on $(0, 1)$.

(i) Use standard linear approximation of f at $x = 0$ to approximate $f(1)$.

(ii) Show that there exist $A \in (-\frac{1}{2}, \frac{1}{2})$ and $c \in (0, 1)$ such that

$$f(1) = f(0) + f' \left(\frac{1}{2} \right) + Af''(c).$$